Inclusive Credit Infrastructure

OPEN-SOURCE DEMO · SYNTHETIC DATA

View on GitHub →

● INTERACTIVE DEMO · ILLUSTRATIVE APPLICANT · SYNTHETIC DATA

Some of your best borrowers don't have a credit file to prove it.

This is a short, open-source demonstration of how traditional credit information can be supplemented with everyday financial signals — rent, utilities, cash flow — to produce an assessment a loan officer can actually read, question, and check for fairness. Not a product pitch. A working example, walked through with one fictional applicant.

Walk through the example ↓

View the code on GitHub

THE IDEA IN THREE PARTS

01

Look beyond the bureau file

With an applicant's consent, weigh rent, utility, and cash-flow patterns alongside — not instead of — traditional credit data.

02

Show the reasoning

Every assessment comes back with plain-language reasons, so a human can see why — not just a score.

03

Check for fairness

Built-in metrics flag when approval rates diverge sharply across groups, so it's monitored rather than assumed.

WALKTHROUGH

Meet the applicant

Six steps, one fictional small business. Each step uses the platform's actual data fields and logic — nothing here is simulated beyond the applicant herself.

Maria's Bakery — working-capital application

FICTIONAL EXAMPLE

1. Applicant profile

Maria has run her bakery for three years. Revenue is steady but seasonal, and — like many small business owners — most of her financial life runs through her business checking account and her landlord, not a credit card.

Business	Maria's Bakery
Structure	Sole proprietor, owner-operated
Time in business	3 years
Requesting	$35,000 working capital

Purpose	Second oven, seasonal demand

2. Traditional credit file

On paper, there isn't much to go on. A thin file like this is common — by some estimates, tens of millions of U.S. adults are considered credit invisible or unscorable using conventional methods alone. It doesn't mean someone is risky. It means the bureau file has little to say.

Credit file status	Thin file
Open trade lines	1 (secured card, 18 mo)
Loan history on file	None
Conventional bureau score	Not available

3. Additional financial signals

With Maria's consent, the platform can also look at day-to-day financial behavior — the kind of information her landlord and utility provider already have, but a credit bureau doesn't.

PAYMENT HISTORY

Rent, on time (24 mo)	100%
Utilities, on time (12 mo)	96%

CASH FLOW (6 MO)

Avg. monthly income	$5,400
Avg. daily balance	$2,150
Volatility	Moderate, seasonal

Account risk events (12 mo)

Overdrafts	0
Non-sufficient-funds events	0

4. Credit assessment

The assessment isn't a yes/no gate — it's a starting point an underwriter can read, question, and combine with their own judgment and policy.

53.7 /100

Favorable — narrowly clears the review threshold

Strong, consistent payment behavior outweighs a thin bureau file — but only just. This is a close call, which is exactly when a human reviewer's judgment matters most. One factor is flagged for a human to weigh — see the next step.

This is the actual output of the platform's baseline model for this scenario — live-scored against the real API, not staged. That model is trained on synthetic demonstration data only, so treat the number itself as illustrative, not a production-grade score. Still not a real underwriting decision.

5. Key factors behind this assessment

Instead of a black-box score, the system returns plain-language reasons — the same kind of information adverse-action notices are meant to convey under fair lending law.

✓ WHAT HELPED

- ✓ Stable, verifiable monthly income

⚠ WORTH WATCHING

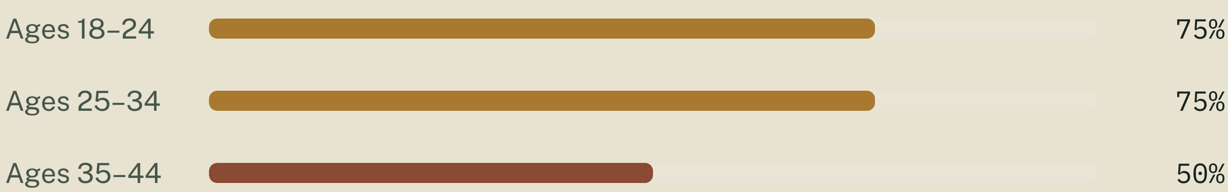
- ⚠ Seasonal cash-flow swings add some repayment uncertainty

- ✓ Regular, on-time utility payments
- ✓ 24 months of on-time rent payments
- ✓ Three years of stable business tenure

Zero overdrafts and zero NSF events in the past 12 months — no other flags raised.

6. Fairness check

This isn't Maria-specific — it's a portfolio-level check the platform runs automatically against a separate synthetic monitoring batch. Below: approval rate by age group, live from that check.



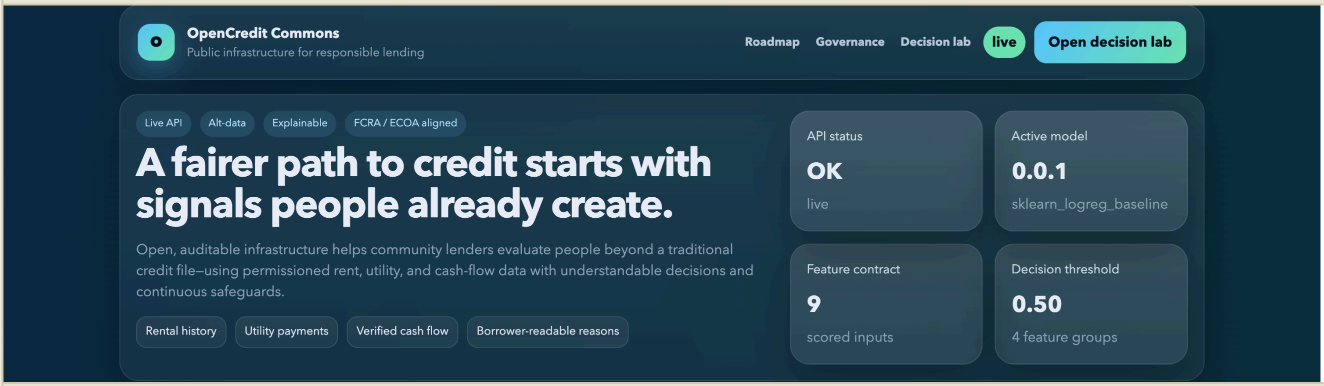
Gap between highest and lowest group: **25 percentage points** — large enough that the platform flagged both fairness metrics for review rather than marking them stable. That's the check working as intended: surfacing a signal for a human to look at, not quietly folding it into an approval rate. It is not proof the model is biased on its own — this batch is a small synthetic sample built to demonstrate the monitor, not a real applicant population.

PROOF, NOT MOCKUPS

This is what's actually running

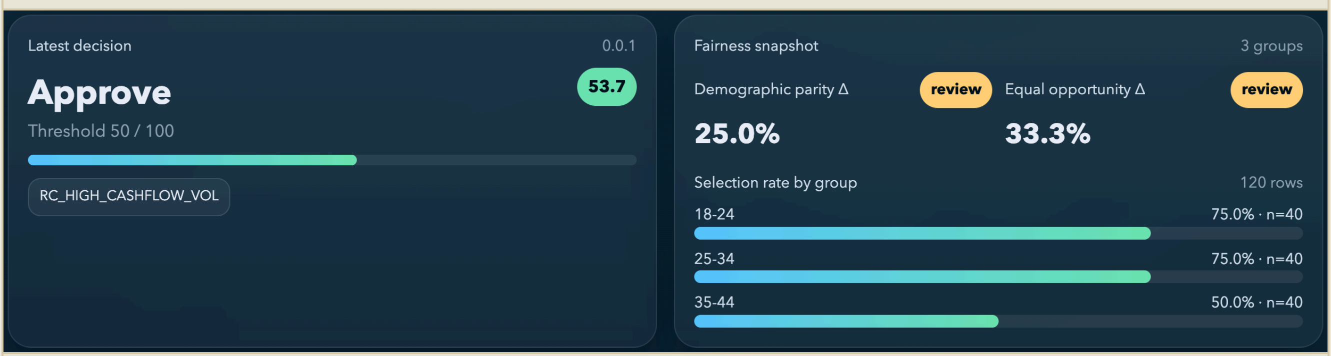
Screenshots below, not illustrations — the open-source reference console, scoring the Maria's Bakery scenario above through the live API and the real trained baseline model.

reference console · local build



The reference console — the same one in this repository, running against a live scoring API.

reference console · decision + fairness

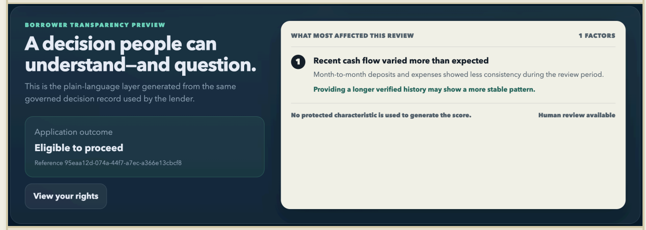


The actual assessment and fairness snapshot for this scenario — not staged.

explainability

The full factor-by-factor breakdown behind the score.

borrower view



The same decision, translated for the applicant.

IN PRACTICE

How a lender might use this alongside existing underwriting

The platform is built to sit next to a lender's process, not replace it.

01

Officer reviews

A loan officer sees the applicant, the assessment, and the plain-language reasons behind it.

02

Weighed against policy

That information is combined with the institution's own underwriting policy, documentation, and judgment.

03

Lender decides

The lending decision — and the compliance responsibility for it — stays with the lender, not the platform.

SCOPE

What this is — and isn't

THIS IS

+ An open-source framework for alternative-data credit assessment
+ A demonstration environment using synthetic data
+ A way to explore explainable, fairness-aware scoring approaches
+ Something a technical team could pilot, test, or extend

THIS IS NOT

– A credit bureau or a production scoring service
– An automatic approve/deny system
– A guarantee of improved lending outcomes
– Currently in use by any financial institution
– Legal or compliance advice

UNDER THE HOOD

The full implementation — feature definitions, the scoring API, fairness metrics, and audit logging — is public.

`Apache-2.0 license · github.com/WangProjects/credit-evaluation-platform`

View repository →